

ISutra — System Architecture & Technical Documentation

AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications

Smart India Hackathon (SIH) Problem Statement SIH26108

1. Executive Summary & Purpose

Public procurement across Government Departments, Public Sector Undertakings (PSUs), and Municipal Corporations in India requires strict adherence to national quality standards. Under **Rule 144(i) of the General Financial Rules (GFR, 2017)** and the **Bureau of Indian Standards Act, 2016**, technical specifications in public tenders must, to the extent practicable, cite active Indian Standards (IS).

In practice, procurement officers face severe operational hurdles:

- **Manual Catalog Inefficiencies:** Searching through 20,000+ BIS standards is slow and error-prone.
- **Tender Obsolescence:** Copy-pasting legacy tender templates leads to citing withdrawn, superseded, or unamended standards, triggering supplier disputes and tender cancellations.
- **Vendor Lock-in & Ambiguity:** Specifications written with proprietary or ambiguous terms restrict competitive bidding.
- **Linguistic Barriers:** Regional procurement tenders written in Indic languages (Hindi, Telugu) cannot easily be mapped to English-standardized national codes.

ISutra solves these systemic challenges. It is an explainable, end-to-end procurement intelligence system that ingests unstructured tender requirements (PDF, DOCX, text, and multilingual queries), extracts structured parameters with provenance tracking, maps them to verified Indian Standards using a **deterministic multi-signal scoring engine**, identifies requirement gaps, verifies lifecycle and amendment history, and outputs an audit-grade **Procurement Intelligence Report** linked directly to official BIS gazette sources.

2. Core Architectural Philosophy: AI Perception vs. Deterministic Decision Logic

A critical flaw in standard GenAI implementations for government workflows is the risk of **LLM hallucination** (e.g., an LLM fabricating standard numbers, inventing safety clauses, or guessing amendment dates).

ISutra implements a strict architectural separation:

AI Perception Layer → Human-in-the-Loop Barrier → Deterministic Decision Engine → Official BIS Ground Truth



3. Detailed Component Architecture

3.1 Frontend (User Experience & Review)

- **Framework:** React 19 SPA powered by Vite 8 and TypeScript.
- **Styling:** Tailwind CSS v4 with custom responsive procurement design tokens.
- **Componentry & Icons:** Lucide React for accessible, institutional UI iconography.
- **Key User Interfaces:**
 - `AnalyzePage.tsx` : Multi-modal tender input (file upload, drag-and-drop, text input, multilingual selector).
 - `RequirementReviewPage.tsx` : Interactive human-in-the-loop review interface allowing officers to verify, edit, and add requirements.
 - `RecommendationsResultsPage.tsx` : Ranked standards listing with relevance scores, specificity badges, and factor breakdowns.
 - `WhyThisStandardPanel1.tsx` : Explainable breakdown showing the exact mathematical contribution of all 6 matching signals.
 - `RequirementGapAnalysisPage.tsx` : Proactive audit of missing vs. supported procurement specifications.
 - `StandardDetailsPage.tsx` & `AlliedStandardsExplorer.tsx` : Normative references, test methods, lifecycle timeline, and gazette amendments.

3.2 Backend (REST API & Intelligence Services)

- **Runtime:** Node.js with Express written in TypeScript.
- **Modularity:** Clean service-repository pattern separating controllers, business logic, and database schemas.
- **Core Micro-Services:**
 - `documentExtractionService.ts` : Real file parsing for `.pdf` and `.docx` with magic-byte signature validation.
 - `aiClient.ts` : Dynamic multi-provider AI client (Gemini 2.5 Flash, OpenAI GPT-4o-mini) with offline deterministic fallback.
 - `multilingualService.ts` : Script heuristic detection and Indic concept translation layer (Hindi, Telugu, English).
 - `standardsMatcher.ts` : Deterministic multi-signal matching engine.
 - `requirementGapAnalyzer.ts` : 4-state parameter gap analyzer.
 - `standardsRelationshipService.ts` : Allied and normative standard relationship resolver.
 - `standardLifecycleService.ts` : Chronological amendment and gazette verification service.
 - `procurementReportService.ts` : Audit-grade printable evaluation report generator.

3.3 Persistence Layer (MongoDB & Curated Dataset)

- **Database:** MongoDB with Mongoose ODM.

- **Connection Resilience:** Centralized safe connection module (`backend/src/config/database.ts`) with URI credential masking, auto-reconnect, and graceful shutdown.
- **Curated Verified Reference Dataset:** Cold-start seeded dataset (`verifiedStandards.ts` , `verifiedStandardLifecycles.ts` , `verifiedStandardAmendments.ts` , `verifiedRegulatoryRecords.ts`) mapping verified BIS records with official source URLs.

4. The 6-Signal Deterministic Matching Engine

The core recommendation engine (`standardsMatcher.ts`) calculates a normalized relevance score ($S \in [0.0, 1.0]$) using a weighted linear combination of six independent domain signals:

$$S = \sum_{i=1}^6 w_i \cdot s_i$$

Factor Weight Distribution:

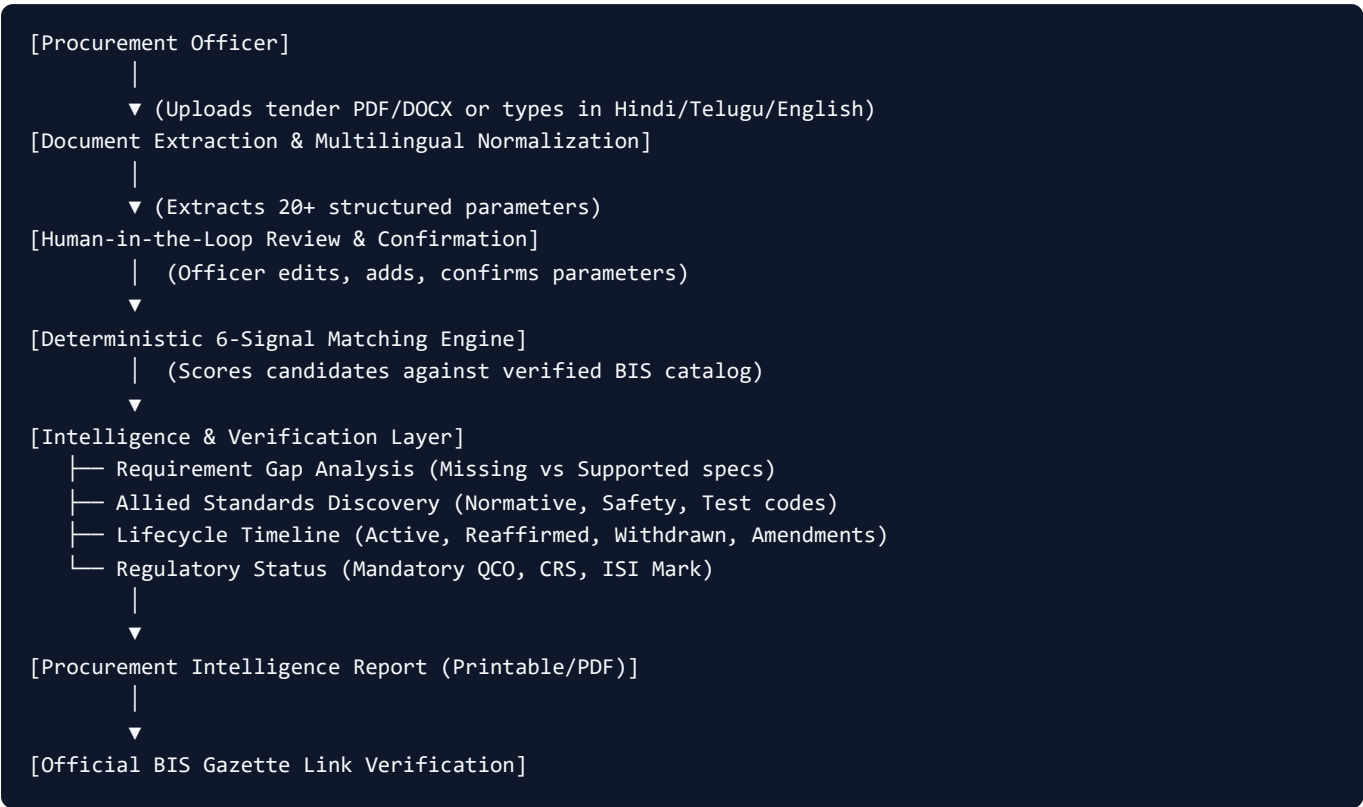
Factor	Weight (\$w_i\$)	Description & Matching Criteria
Product / Category	30% (0.30)	Evaluates direct taxonomy alignment between the confirmed product category and the standard's primary product classification.
Keywords, Title & Scope	25% (0.25)	Matches domain terminology against the standard's official Title and published BIS Scope statement.
Application Domain Fit	15% (0.15)	Evaluates environment/installation suitability (e.g., Highway/Street lighting vs. Industrial vs. Residential).
Environmental Context	10% (0.10)	Matches ambient operating conditions (e.g., outdoor, high-temperature, corrosive/marine, dust ingress).
Technical Parameters	10% (0.10)	Evaluates numerical specifications (e.g., wattage, voltage, lumens/watt, conductor core, steel rebar grade).
Safety & Testing	10% (0.10)	Validates safety protections (surge protection, fire retardance) and mandatory test methods (type tests, routine tests).

Specificity Hierarchy & Penalties

To prevent generic standards from outranking specialized standards:

- **Tiers 1 to 6 Specificity Rating:** Tier 1 represents an exact product-type and application match; Tier 6 represents a broad cross-domain candidate.
- **Product-Form Contradiction Penalties:** When user specifications explicitly demand one application (e.g., *Outdoor Highway Roadway Lighting*), standards explicitly restricted to another form (e.g., *Underground Coal Mining Luminaires*) receive a severe contradiction penalty ($s_i = 0$ with contradiction status flagged).

5. End-to-End Procurement Workflow



6. Verification, Traceability & Trust Models

Government tenders require full auditability before the Central Vigilance Commission (CVC) and the Comptroller and Auditor General (CAG). ISutra enforces trust through:

1. The 5-Stage Audit Traceability Chain

Every recommended standard includes an unbroken chain of custody:

1. **User Input:** Exact clause or paragraph in the uploaded tender document.
2. **Extracted Requirement:** Structured parameter derived during parsing.
3. **Matching Signal:** Specific factor (e.g., Ingress Protection IP66) evaluated by the algorithm.
4. **Standard Data:** Explicit clause or technical table in the verified BIS standard record.
5. **Official BIS Source:** Direct HTTPS link to `bis.gov.in` or `egazette.gov.in`.

2. Four-State Reference Coverage Taxonomy

To avoid false legal claims, ISutra strictly categorizes requirements into:

- `supported` : Confirmed present in the standard.
- `not_supported` : Contradicted or excluded by the standard.
- `not_available` : Standard scope does not govern this specific dimension.

- `needs_verification` : Parameter requires manual inspection in the full BIS publication. (*The system strictly labels this as **Reference Coverage** and never issues a legally binding "compliance certificate"*).

7. Automated Test Suite & Quality Assurance

ISutra is accompanied by six automated test suites containing **over 280 passing test assertions** (0 failures):

Test Suite File	Focus Area	Assertions Verified
<code>backend/test_sih26108_comprehensive.mjs</code>	Full capability suite (Tests A through T)	120+
<code>backend/test_multilingual.mjs</code>	Hindi, Telugu, and English script handling	35+
<code>backend/test_regulatory_certification.mjs</code>	Mandatory QCO, CRS, and Hallmarking logic	30+
<code>backend/test_allied_standards.mjs</code>	Normative references and test method resolution	25+
<code>backend/test_phaseF.mjs</code>	Real PDF/DOCX ingestion and scanned doc handling	30+
<code>backend/test_differentiation.mjs</code>	"Why This Standard?" explainability and scoring weights	40+

8. Deployment Specifications

- Hosting Environment:** Cloud-native deployment on **Vercel** with Express serverless functions.
- Hardware Efficiency: Zero specialized GPU dependency.** The entire deterministic scoring engine, regex parsers, and document extractors run on standard low-cost CPU instances (\$ < 512\$ MB memory footprint).
- High Availability:** Fallback architecture ensures that if MongoDB or external AI APIs are unreachable, the system gracefully operates in offline deterministic mode with zero downtime.

Authored by Team Core Coders / Zero Coders for Smart India Hackathon (SIH) 2026 — Problem Statement SIH26108.